

Technical Report

Comparative Study of Speech-to-Text Models for Noisy Real-World Audio.

Model Research, Benchmarking and Recommendation for Customer Support Applications.

Submitted By: Durwang Deotale.

Email: durwangdeotale34@gmail.com

GitHub: <https://github.com/Durwang2002>

LinkedIn: <https://www.linkedin.com/in/durwang-deotale>

Submitted To: ImmverseAI

Research – Technical (AI/ML)

Assignment 1.

Report Information:

Item	Details
Project Title	Comparative Study of Speech-to-Text Models for Noisy Real-World Audio.
Objective	Benchmark and evaluate state-of-the-art ASR models for customer support applications.
Dataset	LibriSpeech Test-Clean (OpenSLR).
Models Evaluated	OpenAI Whisper Base, Faster-Whisper Base, Facebook Wav2Vec2 Base
Evaluation Metrics	Word Error Rate (WER), Inference Time, Memory Usage, Setup Complexity
Development Environment	Python 3.11, JupyterLab, PyTorch, TensorFlow.
Submission Date	09 July 2026
Report Version	Version 1.0

Declaration:

This report presents a comparative evaluation of Automatic Speech Recognition (ASR) models conducted for the Immverse AI Research – Technical (AI/ML) assignment. All experiments and findings are based on the work performed during this study, with external datasets and references appropriately acknowledged.

Objective:

The objective of this study is to benchmark and compare three Automatic Speech Recognition (ASR) models—OpenAI Whisper Base, Faster-Whisper Base, and Facebook Wav2Vec2 Base—for speech-to-text transcription. The evaluation uses the official LibriSpeech Test-Clean dataset and measures Word Error Rate (WER), inference time, memory usage, and setup complexity under a consistent experimental environment.

The study aims to identify the most suitable model for a voice-based customer support application by analyzing transcription accuracy, computational efficiency, deployment feasibility, and overall production readiness. Based on the benchmark results, practical recommendations, optimization strategies, and a production deployment approach are also proposed.

Objectives of the Study:

- Benchmark three state-of-the-art ASR models.
- Compare transcription accuracy using WER.
- Evaluate inference time and memory usage.
- Assess deployment feasibility and setup complexity.
- Recommend the most suitable model for production use.

Problem Statement:

Automatic Speech Recognition (ASR) plays a critical role in AI-powered applications such as virtual assistants, voice search, and customer support systems. In customer service environments, speech recognition models must accurately transcribe conversations despite challenges such as different accents, speaking styles, background noise, and varying audio quality. Selecting the right ASR model is therefore essential for building reliable and efficient voice-based applications.

Several open-source ASR models are available, each offering different strengths in terms of transcription accuracy, inference speed, resource utilization, and deployment complexity. However, their real-world performance can vary depending on the dataset, hardware, and application requirements. Therefore, a systematic benchmark is required to identify the most suitable model.

This study compares OpenAI Whisper Base, Faster-Whisper Base, and Facebook Wav2Vec2 Base using the official LibriSpeech Test-Clean dataset. All models are evaluated under the same experimental conditions using Word Error Rate (WER), inference time, memory usage, and setup complexity to ensure a fair comparison.

Although LibriSpeech Test-Clean contains high-quality speech and does not fully represent noisy customer support conversations, it provides a standardized benchmark for comparing ASR models. Based on the experimental results and deployment considerations, this study recommends the most appropriate model for production use and proposes future optimization and fine-tuning strategies.

Dataset Description

Dataset Overview:

This study uses the LibriSpeech Test-Clean dataset, a widely recognized benchmark for Automatic Speech Recognition (ASR) research. The dataset consists of high-quality English speech recordings with manually verified transcripts, making it suitable for evaluating transcription accuracy under standardized conditions.

Dataset Source:

- Dataset: LibriSpeech ASR Corpus (Test-Clean)
- Source: OpenSLR
- Language: English
- Audio Format: FLAC
- Transcripts: Human-verified
- License: CC BY 4.0

Dataset Preparation:

The official dataset was downloaded, extracted, and verified before experimentation. Audio files were matched with their corresponding transcripts, and a fixed subset of 30 audio samples was randomly selected using a predefined random seed to ensure reproducibility. Each sample contained the audio ID, audio path, and ground truth transcript.

Dataset Characteristics:

Attribute	Description
Dataset	LibriSpeech Test-Clean
Language	English
Audio Format	FLAC
Benchmark Samples	30
Ground Truth	Human-verified transcripts
Evaluation Purpose	ASR Model Benchmarking

Advantages:

- Official benchmark dataset for ASR research.
- High-quality speech recordings with verified transcripts.
- Standardized dataset for fair model comparison.

Limitations:

The dataset contains clean speech and does not fully represent real-world customer support conversations with background noise, overlapping speech, or telephone audio. Therefore, the benchmark provides a reliable baseline for comparison, while additional evaluation on noisy datasets would further validate production performance.

Summary:

The LibriSpeech Test-Clean dataset provides a consistent and reproducible benchmark for evaluating ASR models. Using the same 30-sample evaluation set for all experiments ensured a fair comparison of transcription accuracy, inference time, memory usage, and deployment suitability.

Research Methodology

Methodology Overview:

A structured benchmarking methodology was followed to ensure a fair, reproducible, and objective comparison of the selected Automatic Speech Recognition (ASR) models. All models were evaluated using the same dataset, experimental environment, and performance metrics to maintain consistency throughout the study.

Research Workflow

The study was conducted in the following phases:

1. Environment Setup
2. Dataset Preparation
3. Model Benchmarking
4. Performance Evaluation
5. Comparative Analysis
6. Model Recommendation
7. Documentation and Reporting

Experimental Environment

The experiments were performed using JupyterLab with Python 3.11 on a Windows system. The required libraries, including PyTorch, TensorFlow, Whisper, Transformers, and JiWER, were verified before execution. Since a GPU was unavailable, all benchmarks were performed on the CPU under identical conditions.

Dataset Preparation

The official LibriSpeech Test-Clean dataset was downloaded and verified. A fixed subset of 30 audio samples was randomly selected using a predefined random seed. Each sample included the audio file path and its corresponding ground truth transcript, ensuring consistent evaluation across all models.

Model Benchmarking:

Three ASR models were benchmarked:

- OpenAI Whisper Base
- Faster-Whisper Base
- Facebook Wav2Vec2 Base

Each model transcribed the same evaluation dataset, and the generated transcripts were stored for further analysis.

Performance Evaluation:

Model performance was evaluated using the following metrics:

- Word Error Rate (WER) for transcription accuracy
- Inference Time to measure processing speed
- Memory Usage to evaluate resource consumption
- Setup Complexity to assess deployment effort

Before calculating WER, both ground truth and predicted transcripts were normalized by converting text to lowercase, removing punctuation, and eliminating extra spaces to ensure a fair comparison.

Methodology Summary:

The adopted methodology ensured a transparent, reproducible, and unbiased evaluation of all selected ASR models. By maintaining identical experimental conditions and standardized evaluation metrics, the study provides a reliable comparison to support evidence-based model selection for real-world deployment.

Models Overview:

Three state-of-the-art Automatic Speech Recognition (ASR) models were selected for benchmarking based on their research relevance, open-source availability, and production applicability.

OpenAI Whisper Base:

Whisper Base is a transformer-based encoder-decoder model developed by OpenAI and trained on approximately 680,000 hours of multilingual speech data. It is released under the MIT License and provides strong transcription accuracy, multilingual support, and robustness across different speaking styles. However, it requires relatively higher computational resources during inference.

Faster-Whisper Base:

Faster-Whisper is an optimized implementation of Whisper built on the CTranslate2 inference engine. It uses the same pretrained Whisper weights while improving inference efficiency and

reducing memory usage. Released under the MIT License, it is well suited for production deployment where computational efficiency is important.

Facebook Wav2Vec2 Base:

Wav2Vec2, developed by Meta AI, is a transformer-based model that uses self-supervised learning to learn speech representations from unlabelled audio before fine-tuning for ASR tasks. It is available under the Apache 2.0 License and offers excellent transcription accuracy with efficient inference. However, domain-specific fine-tuning may be required for optimal performance on noisy real-world speech.

Model Selection Rationale:

These three models represent different ASR approaches and provide a balanced comparison across transcription accuracy, inference speed, memory usage, licensing, deployment complexity, and production readiness, making them suitable candidates for benchmarking in a customer support speech recognition scenario.

Experimental Setup:

The experiments were conducted using JupyterLab with Python 3.11 on a Windows-based system. All required libraries, including PyTorch, TensorFlow, Whisper, Faster-Whisper, Transformers, and JiWER, were installed and verified before benchmarking. Since GPU acceleration was unavailable, all models were evaluated on the CPU under identical hardware conditions to ensure a fair comparison.

The official LibriSpeech Test-Clean dataset was used for evaluation. A fixed subset of 30 audio samples was randomly selected using a predefined random seed, and each model was evaluated using the same dataset without any modifications. Ground truth transcripts were paired with each audio sample to calculate transcription accuracy.

Before computing Word Error Rate (WER), both the ground truth and predicted transcripts were normalized by converting text to lowercase, removing punctuation, and eliminating extra spaces. During benchmarking, WER, inference time, memory usage, and setup complexity were recorded for each model. The collected results were then consolidated for comparative analysis and final recommendation.

Benchmark Results & Analysis:

The three selected ASR models were evaluated on the same 30-sample LibriSpeech Test-Clean dataset using identical experimental conditions. Performance was measured using Word Error Rate (WER), inference time, memory usage, and setup complexity.

Model	Average WER ↓	Inference Time (sec) ↓	Memory Usage (MB) ↓
Whisper Base	0.0569	1.300	3.46
Faster-Whisper Base	0.0643	2.363	0.91
Wav2Vec2 Base	0.0341	1.077	52.90

The benchmark results indicate that Wav2Vec2 Base achieved the lowest Word Error Rate (0.0341) and the fastest inference time (1.077 seconds), demonstrating the best overall transcription performance on the evaluation dataset. Whisper Base delivered competitive accuracy with moderate inference time and stable resource utilization, making it a reliable general-purpose ASR model. Faster-Whisper Base required the least memory but showed slightly lower accuracy and higher inference time in the current CPU-based experimental setup.

Overall, the results suggest that Wav2Vec2 Base provides the best balance of accuracy and speed for this benchmark, while Whisper Base remains a strong choice for multilingual and production-oriented applications.

Recommendation & Conclusion:

Based on the benchmark results, Facebook Wav2Vec2 Base is recommended as the most suitable ASR model for this study. It achieved the lowest Word Error Rate (0.0341) and the fastest inference time (1.077 seconds), demonstrating the best overall transcription performance on the selected evaluation dataset. Although its memory usage was higher than the other models, the improvement in accuracy and processing speed makes it a strong candidate for production environments where transcription quality is a priority.

OpenAI Whisper Base also performed well, providing reliable accuracy with multilingual capabilities and straightforward deployment, making it suitable for applications requiring support for multiple languages. Faster-Whisper Base offered the lowest memory consumption, making it an attractive option for resource-constrained systems, although its accuracy was slightly lower in this benchmark.

Future work can focus on evaluating these models using real-world noisy customer support datasets, fine-tuning them for domain-specific vocabulary, and optimizing inference using GPU acceleration or model quantization. Overall, this study presents a structured and reproducible benchmarking framework that supports informed ASR model selection for practical AI applications.

References:

1. [OpenAI Whisper GitHub Repository](#)
2. [SYSTRAN Faster-Whisper GitHub Repository](#)
3. [Hugging Face – Wav2Vec2 Documentation](#)
4. [LibriSpeech ASR Corpus \(OpenSLR\)](#)
5. [JiWER – Word Error Rate Evaluation Library](#)
6. [PyTorch Documentation](#)
7. [TensorFlow Documentation](#)
8. [Hugging Face Transformers Documentation](#)